MIL-M-40600D

1 February 1973
SUPERSEDING
MIL-M-40600C
5 March 1971

MILITARY SPECIFICATION

MIXER, FLUID, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification covers commercial, electric fluid mixers for counter use.

* 1.2 Classification. Mixers shall be of the following types and styles as specified (see 6.2):

Type 1 - Single spindle
Type 3 - Three spindles
Type 5 - Five spindles

Style 1 - Without paper cup adaptor
Style 2 - With paper cup adaptor

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

LAWS AND REGULATIONS

29 CFR 1910 - Occupational Safety and Health Act (OSHA) Standards

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Copies of specifications, regulations, and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)

Specification A 167-70 - Stainless and Heat Resisting Chromium-Nickel Steel Plate, Sheet and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

National Sanitation Foundation (NSF)

Basic Criteria C-2 - Evaluation of Special Equipment and/or Devices

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, P.O. Box 1468, Ann Arbor, Michigan, 48106.)

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Underwriters' Laboratories, Inc. (UL)

Standard UL 73-1967 - Motor-Operated Appliances.

(Request for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois, 60611; 1285 Walt Whitman Road, Melville, L.I., New York 11746; or 1655 Scott Boulevard, Santa Clara, California 95050.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product.- The electric fluid mixer delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with specification requirements.

* 3.2 Codes and standards. The electric fluid mixer shall comply with the applicable requirements of NSF Basic Criteria No. C-2, UL Standard No. 73, and OSHA Standards 29 CFR 1910, promulgated under the Occupational Safety and Health Act.

* 3.2.1 Compliance. Prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the electric fluid mixer he proposes to furnish under this specification meets the applicable requirements of UL Standard No. 73, NSF Standard No. C-2, and OSHA Standards 29 CFR 1910 as follows:

* 3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the electric fluid mixer has been tested and conforms to UL Standard No. 73.

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* 3.2.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" or display of the NSF seal on the finished electric fluid mixer, or

(2) A certification for the electric fluid mixer issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the electric fluid mixer is being procured, indicating that the electric fluid mixer has been tested and conforms to NSF.

* 3.2.1.3 OSHA. Acceptable evidence of meeting the applicable safety and health standards of OSHA shall be a certificate of compliance, certifying that the electric fluid mixer furnished conforms to OSHA Standards, 29 CFR 1910.

3.3 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial electric fluid mixer provided the completed item complies with all provisions of this specification. Commercial tolerances are applicable to all material thicknesses and diameters.

* 3.3.1 Stainless steel. Stainless steel shall conform to the chemical composition of any type in the 300 series of ASTM specification A 167-70.

* 3.4 Design and construction. The fluid electric mixers shall be designed and constructed as follows: the type 1 shall have a single spindle, the type 3 shall have 3 spindles, and the type 5 shall have 5 spindles. The style 1 shall be furnished without a paper cup adaptor, but with a detachable tapered mixing cup for each spindle. The style 2 shall be furnished with a paper cup adaptor and with a detachable tapered mixing cup for each spindle. The paper cup adaptor shall be adjustable to accomodate all sizes of paper cups normally used with electric fluid mixers.

* 3.4.1 Stand. The stand shall consist of a metal base and vertical section. The vertical section, in addition to supporting the motor and spindle, shall support the metal tapered mixing cup by means of a stainless steel or chrome plated bracket.

* 3.4.2 Mixing cup. The tapered mixing cup shall be made of stainless steel specified in 3.3.1, not less than 0.020-inch thickness, and shall have a minimum capacity of one fluid quart, when tested as specified in 4.4.2.

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* 3.4.3 Motor. Unless otherwise specified (see 6.2), the motor shall be a universal type, suitable for operation on nominal 120-volt alternating-current (A.C.) and direct-current (D.C.) services, not less than 1/7 horsepower for each spindle and not less than 1/2 horsepower for multispindle, single power source types. The motor shall have permanently lubricated bearings.

3.4.4 Electric cord. The mixer shall be provided with a parallel blade, grounding-type, attachment plug cap. The grounding conductor of the cord shall be attached to all exposed dead metal parts of the mixer at one end and the other end to the ground blade wiring terminal of the attachment plug cap. The cord shall be a minimum of 6 feet in length.

* 3.4.5 Switches. The mixer shall be provided with an "ON-OFF" switch operated by placement and removal of the beverage mixing cup. When variable speed motors are provided, the speed shall be adjusted by a manual switch.

* 3.4.6 Fluid flow. The mixer shall provide different speeds of fluid flow through variable speed motor(s) or through interchangeable spindle agitators on single motor, multispindle mixers. All types of agitators for all spindles shall be furnished with each machine.

3.5 Operation. When tested as specified in 4.4.1, the mixer shall operate smoothly, without excessive vibration or noise.

3.6 Finish. All metal surfaces shall be smooth and free from discoloration, rust, and stains. The mixer shall be finished in accordance with the manufacturer's standard practice and shall be free from chipping, cracks, discoloration, holes, runs, sags, thin spots, or any other imperfections.

* 3.7 Marking. Each mixer shall be permanently marked with the manufacturer's name, tradename, model number, or trademark of such known character as to be readily identifiable with the manufacturer.

3.8 Workmanship. The mixer shall be free from dents, punctures, fractures, rough or sharp edges, burrs, slivers, scratches, rough die marks or gouges. All components shall be carefully assembled, properly adjusted and aligned. Threaded fasteners shall not be stripped, fractured, or loose and shall be drawn up tight. The finished mixer shall not be fractured, buckled, punctured or malformed. Chrome-plating shall be free of flakes or blisters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in

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or order, interior packages and shipping containers shall be marked in

the contract or order, the supplier may use his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.2.2 End item inspection. The inspection lot shall consist of all mixers of one type and style offered for inspection at one time.

* 4.2.2.1 Visual examination. The mixers shall be examined for defects listed in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

* TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's standard finish	X	
	Not smooth on all surfaces	X	
	Not free of cracks, chipping, discoloration, holes, thin spots or other imperfections	X	
	Any sags or runs in enamel; flakes or blisters in chrome plating		X
Construction and workmanship (general)	Part missing	X	
	Item not in accordance with manufacturer's standard product except for changes necessary to comply with specification requirements	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general) (cont'd)	Component not as specified	X	
	Paper cup adaptor not adjustable for all cup sizes normally used with blenders (as applicable)		X
	Electric cord not equipped with grounding plug cap	X	
	Part loose, not properly adjusted or aligned		X
	Any rough or sharp edges, burrs, slivers, scratches, rough die marks or gouges	X	
	Any component fractured, buckled, punctured, or malformed	X	
	Binding or excessive play between mating, moving parts	X	
Threaded fasteners	Missing, broken, stripped or fractured		X
	Not drawn up tight		X
Identification marking	Missing, illegible, incomplete, incorrect		X

* 4.2.2.2 Dimensional examination. The mixers shall be in compliance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

* 4.2.2.3 End item testing. The first produced mixer shall be tested as specified in 4.4.2. Any nonconformance shall be cause for rejection. The mixers shall be tested as specified in 4.4.1. The inspection level shall be S-2 with an AQL of 6.5 defects expressed in terms of defects per hundred units.

4.2.3 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking as required by section 5 of this specification are complied with. Defects will be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-4 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

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4.2.3.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closuree, waterproofing, and strapping shall be in accordance with PPP-B-636.

TABLE II. Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Component missing, damaged, or otherwise defective
Contents	Number of mixers per container is more or less than specified

* 4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Tests.

4.4.1 Operation test. The fluid mixer shall be connected to both A.C. and D.C. power source, and operated in each motor speed, when applicable (see 3.4.6) and for each spindle (see 3.4.3), without load to determine compliance with 3.5. Any nonconformance to the operational requirements specified shall constitute failure of this test.

4.4.2 Mixing cup capacity test. The mixing cup shall be filled with water, then emptied into a graduated fluid container to determine compliance with 3.4.2. Inability of the mixing cup to meet the capacity requirement specified shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each mixer, together with mixing cup, and when applicable (see 3.4 and 3.4.6) the paper cup adaptors and agitators, shall be packaged

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in a snug-fitting fiberboard box conforming to style FOL (end opening), type CF, variety SW, class domestic of PPP-B-636. Fiberboard inserts, of the same material as the box, shall be used to immobilize the contents. Vitreous enameled finishes shall be prevented from contact with the fiberboard box or inserts by use of highly calandered or machine-glazed kraft paper. Closure shall be in accordance with the appendix of PPP-B-636.

5.1.2 Level C. The mixers shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Three mixers, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape and reinforced with flat strapping or tape banding in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Three mixers, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

5.2.2.1 When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C. Mixers, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. Special handling markings applicable to fragile items shall apply.

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6. NOTES

6.1 Intended use. The fluid mixer is intended for use in preparing beverages normally served at soda fountains.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type and style required (see 1.2).
(c) Electric motor characteristics if other than specified (see 3.4.3).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7320-0478

Review activity:

Army - MD

User activity:

Army - CE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-M-40600D Mixer, Fluid, Electric

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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